

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

ELECTRONIC SENSORS

(Common to CE, EEE, ME, CSE, IT, FT, AI&DS, CSE(AI), CSE(DS), CSE(AI&ML), CSE(IoT), CSE(CS), CS&D and AI&ML)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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|---|----|
| (a) What do you mean by a passive transducer? | 2M |
| (b) What is the basic principle of a resistive potentiometer? | 2M |
| (c) What are the commonly used materials in Thermal Sensors? | 2M |
| (d) What are the applications of heat flux sensors? | 2M |
| (e) Define Hall effect. | 2M |
| (f) What is the principles of operation of magnetic sensors? | 2M |
| (g) Mention the characteristics of a radiation sensors. | 2M |
| (h) What is Polarization? | 2M |
| (i) List the applications of automation sensors. | 2M |
| (j) Mention the standards for smart sensors. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

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|----|---|-----|
| 2 | (a) What is a Sensor? Write the basic classification of sensors based on characteristics and order. | 5M |
| | (b) Explain the static characteristics of a sensor. | 5M |
| | OR | |
| 3 | (a) Explain the resistance strain guage in detail. | 5M |
| | (b) Explain the Sensitivity and types in Inductive Sensors. | 5M |
| 4 | (a) List various gas thermometric sensors. Explain piezoelectric gas sensor. | 5M |
| | (b) Describe in detail Optical gas sensor. | 5M |
| | OR | |
| 5 | Explain the following:
(i) Acoustic Temperature Sensor,
(ii) Thermo emf Sensors. | 10M |
| 6 | (a) Explain the principle of operation of the Magneto-resistive sensors. | 5M |
| | (b) Explain the principle of Semiconductor Magneto resistor. | 5M |
| | OR | |
| 7 | With neat diagram explain the Angular/Rotary movement Transducer. | 10M |
| 8 | (a) Write in detail about the Electro chemical cell. | 5M |
| | (b) Write in detail about Standard Hydrogen Electrode. | 5M |
| | OR | |
| 9 | (a) Explain the working of a fibre optic sensors. | 5M |
| | (b) Explain the working of Nuclear radiation sensors. | 5M |
| 10 | Explain in detail the home appliance sensors with any example. | 10M |
| | OR | |
| 11 | Explain clearly about Aerospace Sensor. | 10M |
